



Better Care, Sooner

Predict Future Risk and Rising Cost

Significant Healthcare AI Spin-out from Stanford Start-X



4 Years
R&D



21 Million
Lives



2.5x
Accuracy



3
Products



21
Advisors and Board



28
Employees



2
Customers Paid



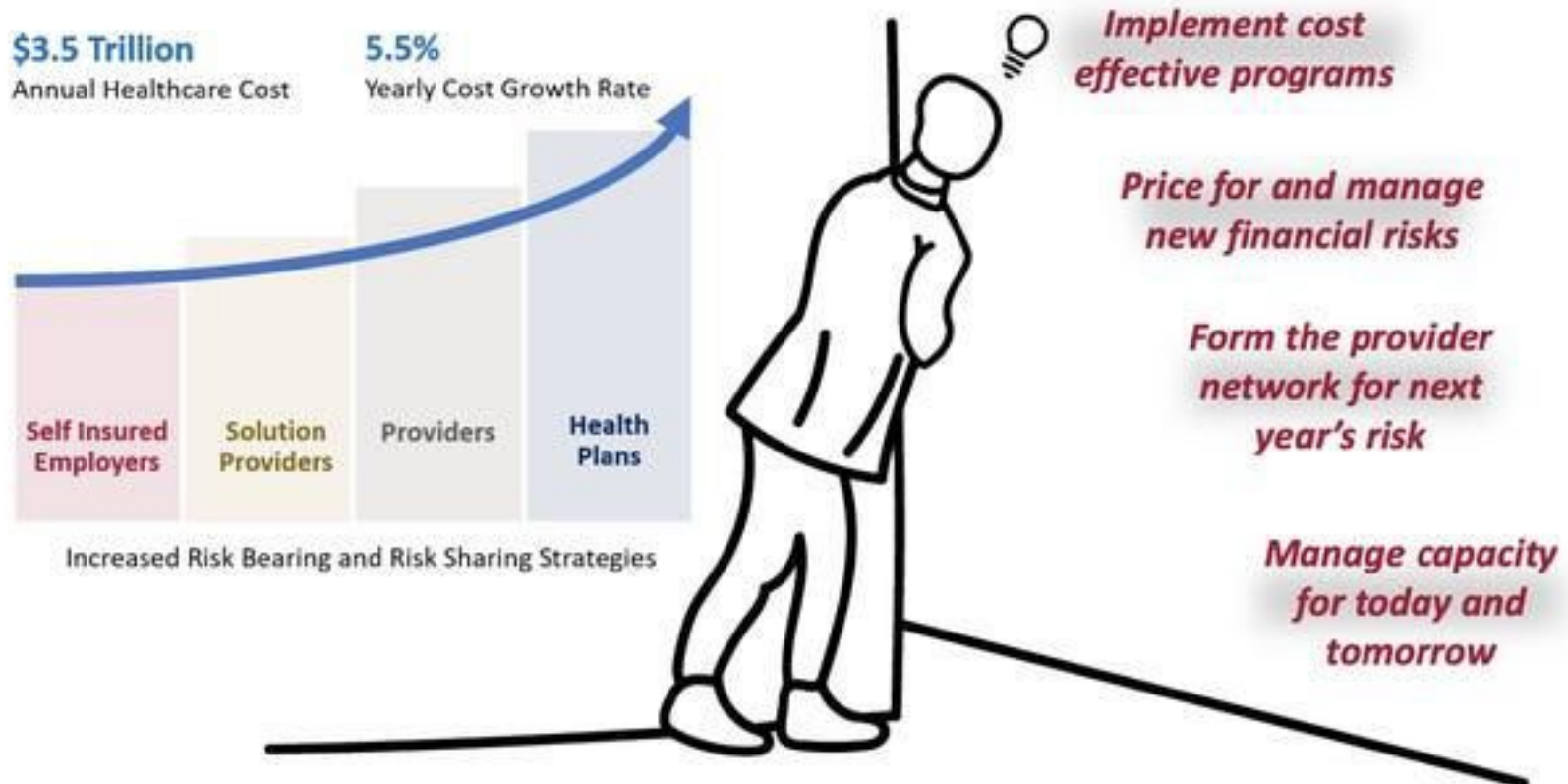
8
Customers MSAs



20
Pipeline Customers

Predicting Future Risk and Rising Cost

6% of today's population will account for 1/3 of next year's new high cost



Predict the What, Why and When of Tomorrow's Cost

Mitigate Rising Risk

Solution 1: COST BLOOM INTERVENTION

- Cost Bloom, Jump, High Cost Claimant Likelihood
- Engagement Likelihood
- Clinical Impactability Score
- Case Selection Optimizer
- Action Plan Recommendation

Ave Annual Price Per
Customer: **\$850k**

Steer High Cost Events

Solution 2: STEERAGE PRECISION

- Clinical Event Likelihood
- Engagement Likelihood
- Event Timing Likelihood
- Case Selection Optimizer
- Case Summary

Ave Annual Price Per
Customer: **\$400k**

Predict Total Cost

Solution 3: RISK ASSESSMENT

- Prospective Risk Score & Cost
- Concurrent Risk Score & Cost
- Cost Bloom, Jump, High Cost Claimant Likelihood

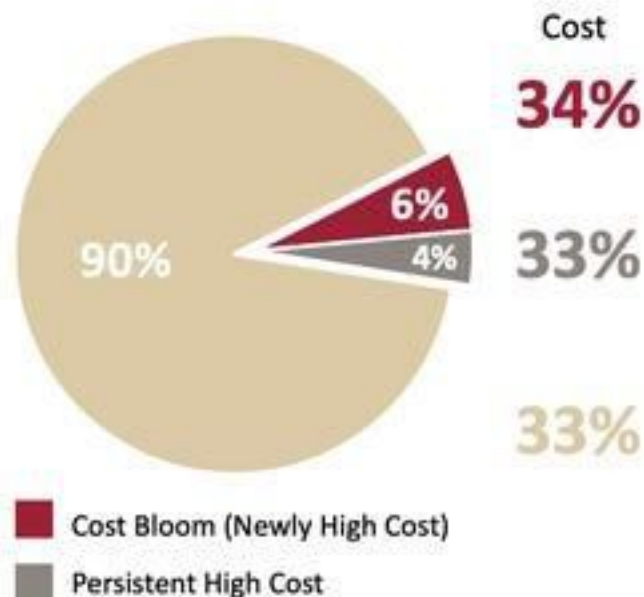
Ave Annual Price Per
Customer: **\$600k**

8x ROI on Highest Impact Cost Blooms

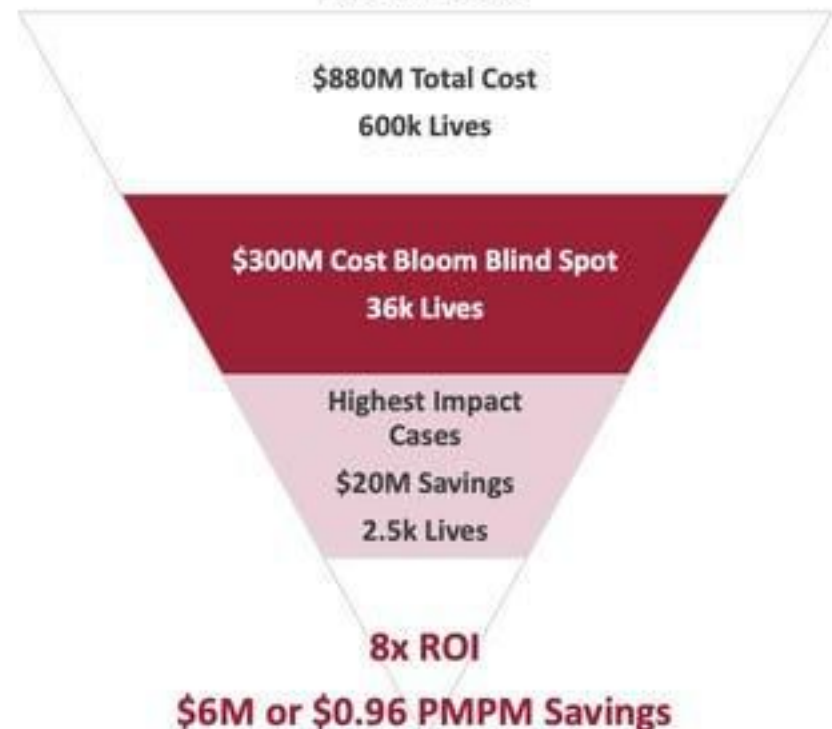
"I felt like it was meant to be for me to reach this member, it was the perfect moment in time"
— Care Manager

Patient Outcome: 57 year old male with heart conditions, back pain, and opioid dependence was steered to a higher quality lower cost spinal fusion center, a savings of over \$20k.

Health Plan A Population



Health Plan A



Cost Bloom Adoption at a Blues Plan



40% engagement rate (one call with Cost Bloom member)



26% active engagement rate (multiple calls with member)



75% successful intervention rate for actively engaged

6x ROI on Ortho Joint Surgery Steerage



1/3

Of cost blooms were ortho joint surgeries



\$3.6M savings, 6x ROI

Projected Ortho Joint savings at Health Plan A

Steerage Precision Addressing a Market Gap



Over \$5M per year in Total Savings

Even larger market beyond Ortho Joint: Bariatric, High Risk OB, Hysterectomy, etc.

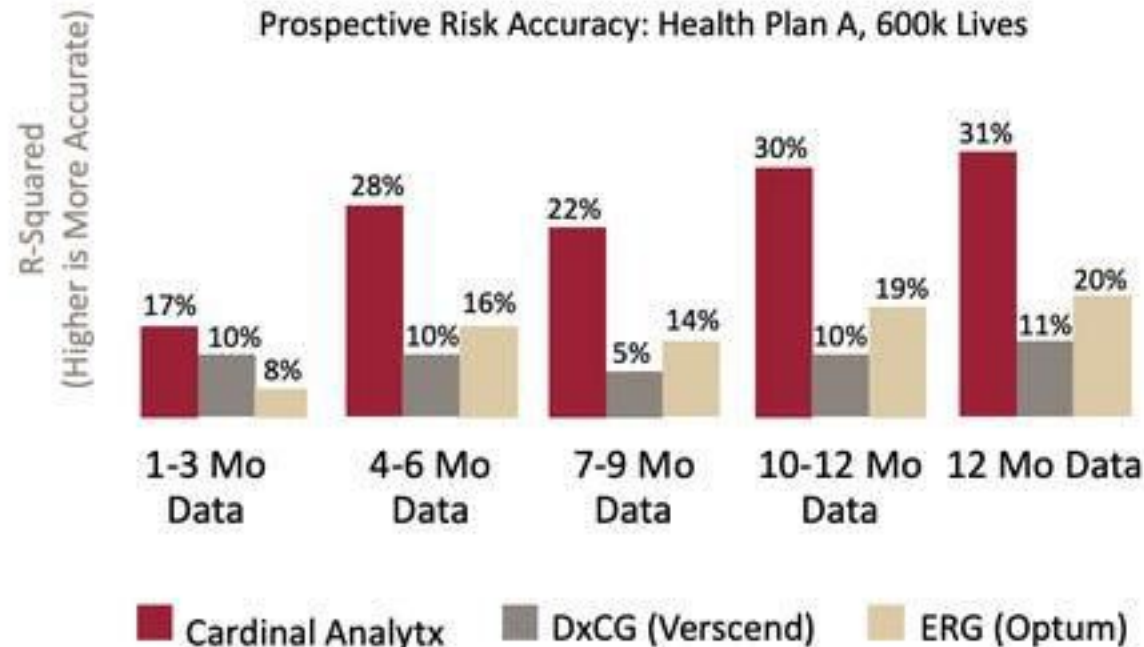


50% of Cardinal Analytix Clients Expressed Interest in Steerage

Cardinal Analytix is working with solution providers to partner (e.g., Vitals, Relay, MOBE, VIM)

\$100M Through Increased Accuracy

Our proprietary Machine Learning beats the competition



Missed Dollars by Competitors:

\$63M missed dollars
Under-pricing risk

\$37M missed dollars
Over-pricing / retention risk

Strong Competitive Position

Strategic Approaches	Competitive Landscape	
1. Machine Learning Predictions with Drivers and Actions	 KenSCI (Provider) Cyft (CM platform)	
2. Machine Learning Predictions + Model Correlations	Base Health (Med Adv. Risk Adj) Jvion (Provider) Ayasdi (Provider)	
3. Machine Learning Predictions	Lumiata (Risk) Cogitativo (Payment) HBI Solutions (Risk) IBM Watson (Truven)	Ark.one (Risk) NextHealth (Provider) Hindsait (Data Aggregation)
4. Traditional Predictions	Optum (Risk) Milliman (Risk) Cave Consulting (Risk) Verscend (Risk) Lexis Nexis (Risk) Deerwalk (Risk)	Health Catalyst (Provider) Welltok (Member Platform) Lightbeam (Provider, Risk) HCC (Risk) ACG (Risk)

Professional Team



Linda Hand
CEO

- 35 years in organizational leadership, product development, solutions delivery and go-to-market strategies
- Successful exit of DecisionView to IMS Health, Clinical Tech core of IQVIA / Quintiles merger
- BA in Computer Science, UC Berkeley, Haas School of Business Executive Program



Niall O'Cathasaigh
Chief Financial Officer

- 20 years of startup experience across 15 venture backed startups, 4 Healthcare companies. 7 years as founder and fund manager. 3 years at Apple.
- CIMA Qualified. DBS from University College Dublin. BA in History from University College Galway and Universite De Poitiers.



Brian Maples
VP of Data Science

- 15 years in biomedical R&D and machine learning
- Invented and patented molecular diagnostic tech at Ionian, acquired by Alere to develop CLIA waived rapid flu test; Led Nuna's data science development of a Network Solutions product; at Stanford, created now leading ancestry inference methodologies
- PhD in Biomedical Informatics from Stanford University, MS in Statistics from Stanford University, BA in Science and Management from Claremont McKenna College



Lu Lu
Head of Product

- 12 years healthcare experience in strategic and clinical transformation initiatives focused on transition to value with over 19 Health Systems, 45 clinical departments, and 450 design participants.
- MPH from Harvard School of Public Health, BA in Psychology from UCLA.



Chris DeRienzo
Chief Medical Officer

- 11 years clinical experience, double-boarded MD
- While serving as Mission Health Chief Quality Officer gained >\$65M through continuous improvement, pay for value and population health initiatives.
- Spearheaded integration of analytics and data science while reducing harm and scaling clinical pathway development.
- MD/MPP from Duke University, BS in Biology, Biochemistry, and Psychology from Brandeis University

Engaged Founders, Investors & Board

Founders



Nigam Shah

Associate Professor of Medicine,
Stanford

- Associate Professor of Medicine, Funded by 5 NIH Institutes, FDA and 7 companies; Asst Director of Center for Biomedical Informatics Research.
- Fellow, American College of Medical Informatics and American Society for Clinical Investigation.
- MS and MD Baroda College (Baroda, India), PhD, Penn State, Postdoctoral, Medical Informatics, Stanford.



Arnold Milstein

Professor of Medicine,
Stanford

- Professor of Medicine and Director, Stanford Clinical Excellence Research Center; Elected Member Institute of Medicine Section of National Academy of Sciences; Medical Director at Pacific Business Group on Health; Board Trustee Intermountain Healthcare.
- Founder national health care performance improvement firm and Leapfrog Group; former Congressional MedPac Commissioner.
- BA, Harvard; MD, Tufts; MPH, U.C. Berkeley.

Investors



**John Doerr
Family Fund**



Board



Thomas McKinley

Chairman of Cardinal Analytix,
Partner at Cardinal Partners



Linda Hand

CEO of Cardinal Analytix



John Doerr

Chairman, Kleiner Perkins



Mark Smith

California HealthCare Foundation,
Co-leads a CMS effort to
accelerate value-based payment



Nigam Shah

Associate Professor of Medicine,
Stanford



Elizabeth Spaulding

Partner at Bain & Company,
Global Head of Bain Digital



Kent Marquardt

Former EVP and CFO at Premera
Blue Cross

Industry Leading Advisors

Science



Susan Athey, PhD
Economics of Technology
Professor, Stanford
Graduate School of Business



Trevor Hastie
The John A. Overdeck
Professor of Mathematical
Sciences, Stanford University



Colt Courtright, MPA
Director, Corporate Data
& Analytics

Industry



Virginia McFerran, MBA
Formerly, Partner, Optum
Ventures, President and
CEO, Optum Analytics



Ray Bradford, MBA
Founder and CEO,
Spruce Health

Health Plan



Jeffrey Bailet, MD, MSPH
Executive VP of Health Care
Quality and Affordability
Blue Shield of California



David Fields
President
Dean Health Plan



**John Espinola, MD,
MPH, MBA**
Executive VP of Healthcare
Services, Premiera Blue Cross



Juan Chaluja
VP Commercial Markets
Analytics and Business
Performance, Florida Blue

Clinical



Kevin A. Schulman, MD, MBA
Professor of Medicine,
Assoc Chair, Bus Dev & Strategy,
Stanford School of Medicine



Ken Paulus, MA
Past President and CEO,
Allina and Atrius Health Systems



Ronald A. Paulus, MD
President and CEO,
Mission Health



Lee Sacks, MD
Former CMO, Advocate
Aurora Health



Alan Glaseroff, MD
Faculty, Stanford
University's Clinical
Excellence Research Center

Investment Opportunity

